

NAME _____

ECS401: Unit 8 Writing programs Progress Test: Arrays of Records

A program has been partly written to input and store the details of cars, based on a record with fields recording their registration plate and make of car. The partial program is given below.

Complete the program by writing the TWO missing methods:

1. Write a createCar method that creates the record, allocating space and setting values for both registration and make. These values should be provided as arguments to the method.
2. Write an inputCars method that inputs the details of 50 cars into a given array of Car records.

```
// A program to input and print car details
// AUTHOR Paul Curzon
// V2
// 13 December 2020

import java.util.Scanner;

class cars
{
    public static void main(String []p)
    {
        Car[] cars = new Car[50];
        inputCars(cars);                // This method needs writing
        printCars(cars);
        System.exit(0);
    }

    // Input and store the details of 50 cars
    //

    // Print the details of all the cars stored
    //
    public static void printCars(Car [] c)
    {
        for(int i=0; i<c.length; i++)
        {
            System.out.println(carToString(c[i]));
        }
        return;
    }

    // input a string given a message
    public static String input(String message)
    {
        Scanner scanner = new Scanner(System.in);
        System.out.print(message + " ");
        return scanner.nextLine();
    }
}
```

```
/* An ADT for type Car.  
   It has the externally usable operations  
   - createCar: create a Car value given the registration and make  
   - carToString: give a printable version of a Car value  
*/
```

```
// get the make of the car  
//  
public static String carToString (Car car)  
{  
    String car_to_print = getCarMake(car) + ": " + getCarReg(car);  
    return car_to_print;  
}
```

```
// create a car record with registration and year  
//
```

```
// get the make of the car (not used outside ADT)  
//  
public static String getCarMake (Car c)  
{  
    return c.make;  
}
```

```
// get the registration of the car (not used outside ADT)  
//  
public static String getCarReg (Car c)  
{  
    return c.registration;  
}
```

```
}
```

```
class Car          // Car details are a registration plate number and the car make  
{  
    String registration;  
    String make;  
}
```